

AI-Powered Smart Attendance Web Application

Gayatri Mahajan (34)

Manya Naik (40)

Guide: Prof. Mr. Bharat Patil Sir

Usha Mittal Institute Of Technology

Technical Seminar: Phase I

Introduction

- Manual attendance is time-consuming, error-prone, and allows proxy marking.
- **Aim:** Secure, smart attendance system.
- Uses **Face Recognition + Wi-Fi + Lecture Token**.
- Ensures only physically present students are marked.

Literature Survey

- **Gaps in Existing Systems (e.g., Smitha et al., 2020):**
- **Low Security:** No mechanism against proxy marking outside the classroom.
- **Data Handling:** Systems often use flat files (Excel/CSV).
- **Accuracy:** Reliance on simpler algorithms (LBPH).
- Many systems use phone camera → **proxy possible**.
- Some use Wi-Fi or RFID → face not verified.
- **Our approach:**
- Multi-Layer Security **face + Wi-Fi + auto token**.
- **Anti-Proxy Design** : Students cannot mark attendance outside class, even if on Wi-Fi.
- Token ensures time-bound lecture verification.

Proposed System Architecture

Hardware Design:

- Classroom Wi-Fi
- Student Smartphone

Software Design:

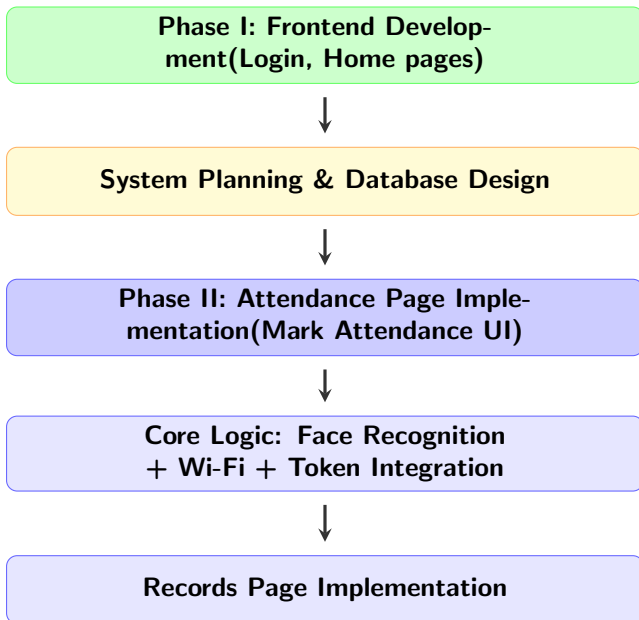
- **Frontend:** HTML, CSS, JS (Login, Home, Mark Attendance, Records)
- **Backend:** Python (Face Recognition, Wi-Fi check, Token verification)
- **Database:** Student records and attendance

Implementation

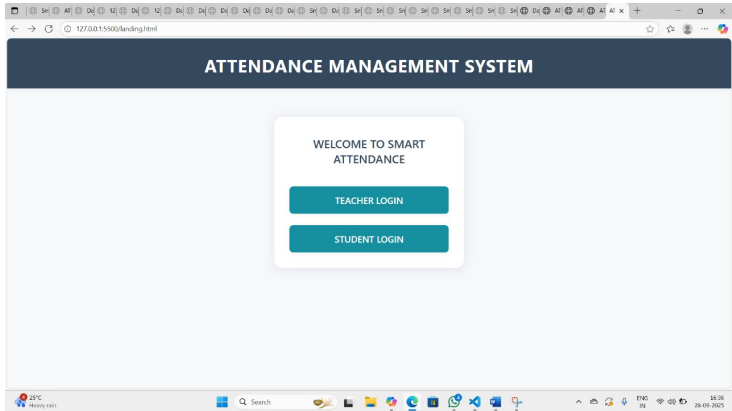
- **Frontend:** Login page (Student/Teacher), Home page, Attendance page.
- **Backend:**
 - Face Recognition using Python
 - Wi-Fi connection check
 - Lecture token validation (auto-generated)
- **Teacher Module:** Manage students, view all records
- **Student Module:**
 - Check attendance status
 - Mark attendance with security checks

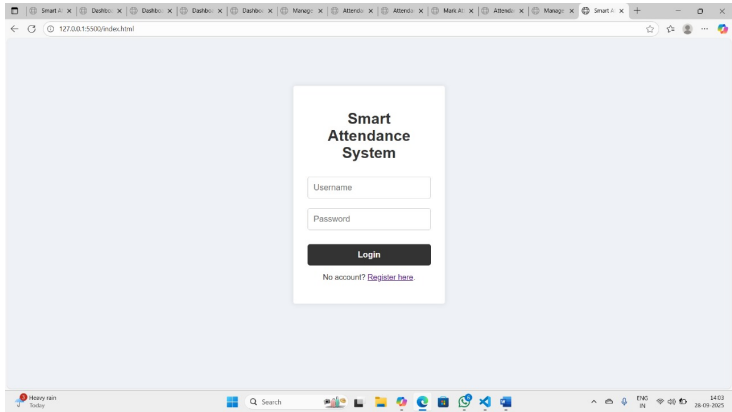
- Frontend design successfully completed.
- Pages developed: Login (Student/Teacher), Home, Attendance.
- Navigation between pages working smoothly.
- Layout is responsive, user-friendly, and professional.

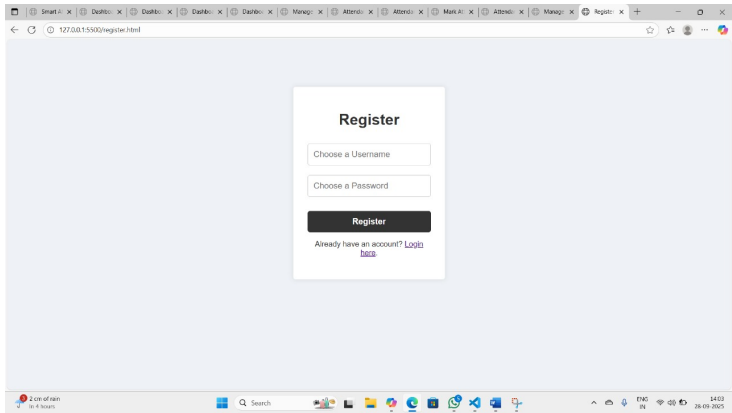
Project Road map

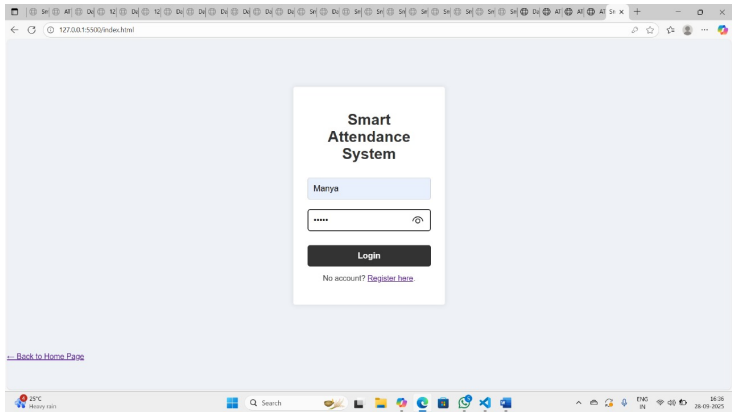


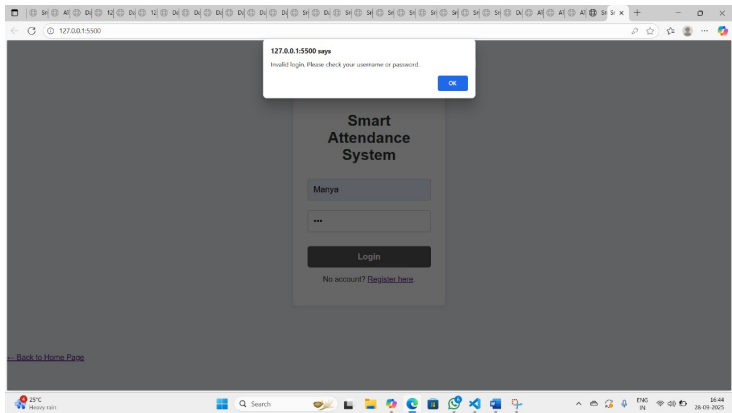
Web Interface

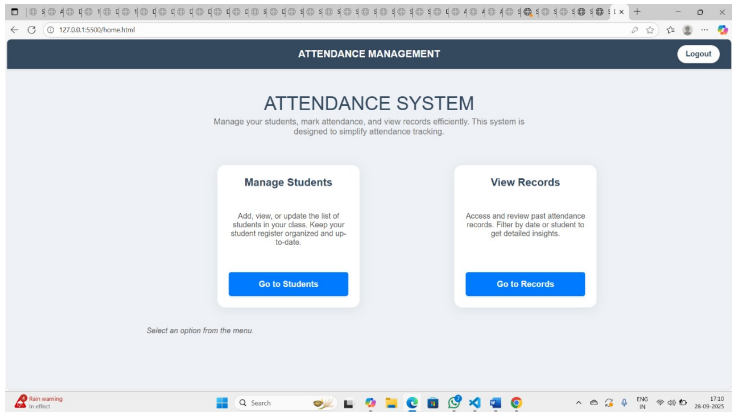


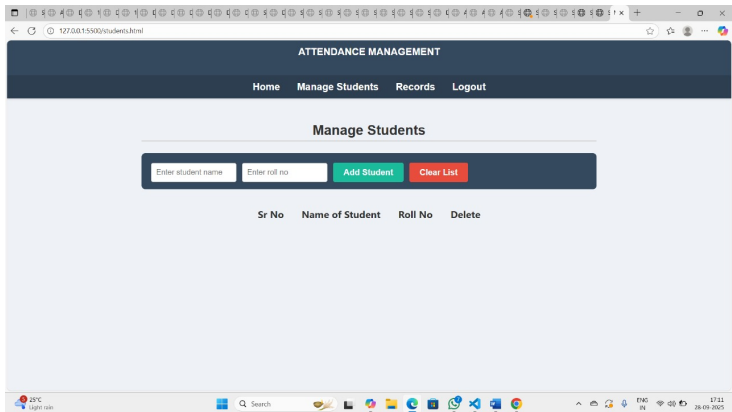












Manage Students - Smart Attendance System

127.0.0.1:5500/students.html

ATTENDANCE MANAGEMENT

Home Manage Students Records Logout

Manage Students

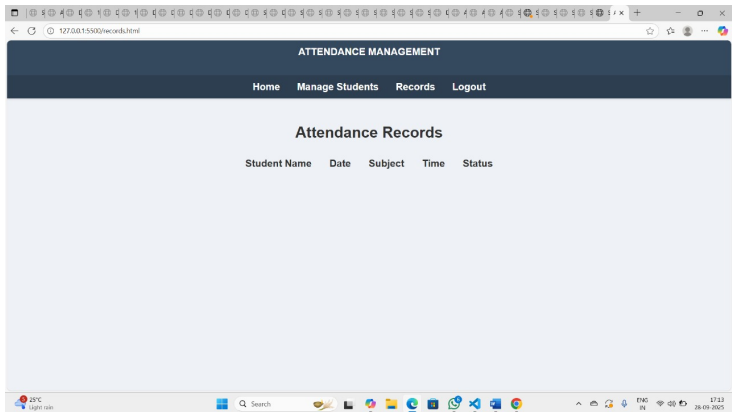
Enter student name Enter roll no Add Student Clear List

Sr No	Name of Student	Roll No	Delete
1	MANYA NAIK	40	Delete
2	GAYATRI MAHAJAN	40	Delete

Train warning in effect

Search

17:40 28-09-2025



Thank You!